12TH CHEMISTRY · CHAPTER 27 · SOLVED EXERCISE

Biochemistry

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. Multiple Choice Questions:

I. Glucose is a building block for all, except:

(a) cellulose (b) glycogen
(c) inulin (d) starch

Answer: (c) inulin — Cellulose, glycogen and starch are polymers of glucose; inulin is built from fructose units.

Inulin is not discussed in the chapter; the fact that it is a polymer of fructose is general knowledge.

II. Which one of the following is not an aldose?

(a) glucose (b) galactose
(c) mannose (d) fructose

Answer: (d) fructose — Fructose contains a ketonic group, so it is a ketose.

III. Which one of the following element is not present in all proteins?

(a) carbon (b) hydrogen
(c) nitrogen (d) sulphur

Answer: (d) sulphur — Sulphur is present only in proteins which contain sulphur-containing amino acids.

IV. In DNA, the linkages between different nitrogenous bases are:

(a) phosphate linkage (b) hydrogen bonding
(c) glycosidic linkage (d) peptide linkage

Answer: (b) hydrogen bonding — Two hydrogen bonds join A to T and three join G to C.

V. Which one of the following nitrogenous bases is not common to RNA and DNA?

(a) cytosine (b) adenine
(c) thiamine (d) uracil

Answer: (d) uracil — Uracil is found only in RNA (bases A, U, G, C).

Option (c) is printed "thiamine" (which is vitamin B_1); if it is read as thymine, then thymine (DNA only) and uracil (RNA only) are both not common to DNA and RNA. Uracil is the option that is correctly named.

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VI. Waxes are the esters of:

(a) glycerols (b) sterols
(c) ethanol (d) long-chain fatty acids and alcohols.

Answer: (d) long-chain fatty acids and alcohols. — Long fatty acid chains are esterified to long-chain alcohols.

VII. A cell membrane is made up of:

(a) phospholipid bilayer (b) lipid
(c) phospholipid (d) only protein

Answer: (a) phospholipid bilayer — The bilayer forms the structure of all cell membranes.

VIII. The bonds in protein structure that are not broken on denaturation.

(a) hydrogen bonds (b) peptide bonds
(c) ionic bond (d) disulfide bonds

Answer: (b) peptide bonds — Denaturation disrupts the folding; the primary sequence held by peptide bonds remains.

IX. The backbone of nucleic acid structure is constructed by:

(a) peptide bonds (b) glycosidic bonds
(c) phosphodiester bridges (d) all of them

Answer: (c) phosphodiester bridges — Sugars are joined through phosphate groups by phosphodiester linkages.

X. Which element prevents the development of tooth decay:

(a) fluorine (b) calcium
(c) phosphorus (d) sodium

Answer: (a) fluorine — Fluoride hardens the enamel of teeth.

The chapter only lists fluoride among the trace minerals; its role against tooth decay is general knowledge.

Q2. SHORT ANSWER QUESTIONS:

a) What role does glycogen play in the human body?

Glycogen (‘animal starch’) is a polymer of glucose stored in the liver and muscles. Excess glucose is stored in this form, and it serves as the energy reserve which supplies glucose for the body and for sustained muscular activity.

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b) Point out the main structural difference between the compounds in each of the following pairs. i) Glucose and fructose ii) Cellulose and starch iii) Competetive and non-competitive inhibition iv) Glycolipids and lipoprotein

Pair	First	Second
i) Glucose / fructose	An aldose (aldehydic group); forms a six-membered ring (pyranose).	A ketose (ketonic group); forms a five-membered ring (furanose).
ii) Cellulose / starch	Long unbranched chains of glucose held together by strong hydrogen bonds; rigid and insoluble.	Made of amylose (unbranched) and amylopectin (compact, highly branched).
iii) Competitive / non-competitive inhibition	The inhibitor has the same shape as the substrate and occupies the active site (reversible).	The inhibitor changes the shape of the active site (irreversible).
iv) Glycolipids / lipoproteins	Lipids with carbohydrate (sugar) groups attached.	Complexes of lipids with proteins.

c) Which class of biomolecule contains each of the following linkages? Describe the function of each linkage. i) Glycosidic ii) Peptide iii) Ester iv) Phosphodiester

Linkage	Class of biomolecule	Function
i) Glycosidic (C–O–C)	Carbohydrates	Joins monosaccharide units into disaccharides and polysaccharides.
ii) Peptide (–CO–NH–)	Proteins	Joins amino acids into polypeptide chains.
iii) Ester (–COO–)	Lipids	Joins fatty acids to glycerol (or other alcohols) in triglycerides, waxes and phospholipids.
iv) Phosphodiester	Nucleic acids	Joins nucleotides through the 5′ and 3′ carbons of the sugars; gives strength and stability to the chain.

d) How the tertiary structure of proteins is formed?

It is formed when a polypeptide chain folds due to interactions between its side chains (R-groups): hydrogen bonds, ionic bonds, hydrophobic interactions and disulphide bridges. These make the chain twist and bend into a specific three-dimensional shape, often a globular protein.

e) In what way fats and oils are different?

Fats	Oils
Contain saturated fatty acids (no C=C double bonds).	Contain unsaturated fatty acids with one or more C=C double bonds.
Solid at room temperature.	Liquid at room temperature.
Mostly of animal origin (animal fats).	Mostly of plant origin (vegetable oils).

f) Describe the structure of steroids. Write a note on the functions of cholesterol.

Structure: steroids consist of three cyclohexane rings and one cyclopentane ring fused together, called the steroid nucleus; the rings are named A, B, C and D. Many steroids also have an –OH group (sterols).

Functions of cholesterol: cholesterol is a steroid (sterol) of the body. Steroid hormones, bile salts and vitamin D belong to the same family and are made from it; it is also a component of cell membranes. It is carried in the bloodstream by lipoproteins.

The chapter names cholesterol among the steroids but does not list its functions separately; the points on cell membranes and on its conversion into hormones, bile salts and vitamin D are general knowledge.

g) Describe the mechanism of enzyme action.

An enzyme has a special site on its surface, called the **active site**, which binds the substrate molecule and converts it into the products. The random movement of enzyme and substrate molecules allows the substrate to enter the active site. A temporary **enzyme–substrate complex** is formed. After this, the products are released and the enzyme remains unchanged, free to bind another substrate molecule.

$$\text{Enzyme} + \text{Substrate} \rightleftharpoons \text{Enzyme–substrate complex} \rightarrow \text{Enzyme} + \text{Products}$$

Lock and key model (Emil Fischer, 1894): the lock is the enzyme and the key is the substrate. Only the correctly shaped key (substrate) fits into the keyhole (active site) of the lock (enzyme).

h) Why does an enzyme catalyze a reaction of only a specific substrate?

Enzymes are highly specific: each enzyme acts on only one particular substrate. This specificity is due to the active site of the enzyme, which has a unique shape that allows only a specific substrate to fit into it, like a key in a lock.

i) Name different RNAs and discuss their function.

Type of RNA	Function
Messenger RNA (mRNA)	Carries the genetic blueprint copied from the DNA in the nucleus.
Ribosomal RNA (rRNA)	Helps to make up the ribosome.
Transfer RNA (tRNA)	Translates the nucleotide sequence in mRNA into the corresponding sequence of amino acids in a protein.

j) What are some sources of potassium, iron, calcium, magnesium and zinc in our diet?

Mineral	Sources
Potassium	Salt, meat, seafood, fruit juices, vegetables, lentils (given with sodium and chloride)
Iron	Liver, meat, egg yolks, nuts, enriched or whole grains, beans, peas and lentils
Calcium	Milk and other dairy products, greens, broccoli, salmon, sardines, beans, peas and lentils
Magnesium	Green leafy vegetables, nuts, whole grains and legumes
Zinc	Meat, seafood, whole grains

Table 27.1 gives no sources for magnesium; that row is from general knowledge.

k) What is the role of iron in the body?

Iron is needed for the synthesis of haemoglobin in red blood cells and myoglobin in muscle cells; it is needed to carry oxygen. In haemoglobin the iron atom attaches to oxygen as the blood travels between the lungs and the tissues. It is stored by the protein ferritin.

l) How enzyme inhibitors are important in our body and medicine?

Inhibitors stop the catalytic activity of enzymes, so they can be used to control or block particular reactions. In medicine, many drugs and antibiotics are inhibitors: penicillin irreversibly blocks the enzyme trans-peptidase, which bacteria need to build their cell wall, and so kills bacteria. Poisons also act as irreversible inhibitors, which is why they can cause severe health issues or death.

Q3. Construction Response Questions

a. Why does the digestive enzyme pepsin have an optimum pH of 2?

Every enzyme works best at its optimum pH, at which its active site has the correct shape. Pepsin works in the stomach, which is strongly acidic because of hydrochloric acid (pH about 2). Its structure is suited to this acidic medium, so its activity is maximum at pH 2; at other pH values its shape is disturbed and it is deactivated.

Pepsin is not discussed in the chapter; the acidic medium of the stomach is general knowledge.

b. How condensation of sugars and condensation of amino acid are same?

In both, two units join through a covalent bond with the release of a molecule of water. Two monosaccharides join through a glycosidic bond (C–O–C); two amino acids join through a peptide bond (–CO–NH–). Repeating the process gives polymers: polysaccharides and polypeptides (proteins).

c. Why is cholesterol included in the lipid family?

Cholesterol is a steroid. Steroids are not formed from fatty acids, but they have lipid-like properties: they are hydrophobic and soluble in organic solvents. They are therefore classed as derived (associated) lipids.

d. Why fats and oils are also called triacylglycerols?

Because they are esters formed from one glycerol molecule and three fatty acids: three acyl groups (R–CO–) are attached to glycerol through three ester links. Hence the names triglycerides or triacylglycerols.

e. How hydrogen bonding is important in the double helix of DNA?

Hydrogen bonds between the bases hold the two polynucleotide chains together in the double helix. They form only between specific pairs: two between A and T and three between G and C, which ensures accurate base pairing and copying of the genetic information. Because they are weak, the strands can separate for replication and transcription.

f. If the sequence T-A-C-C-G-A located on one strand of DNA, what sequence would be seen opposite to it on the other strand?

A pairs with T and G pairs with C, so the opposite strand is:

T–A–C–C–G–A

A–T–G–G–C–T

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DESCRIPTIVE QUESTIONS

Q4. Classify and describe the structure of polysaccharides.

Polysaccharides are large molecules consisting of a large number of monosaccharide units joined together. They have high molecular mass and the general formula $(C_6H_{10}O_5)_n$. On hydrolysis they give simple sugars (monosaccharides). Common examples are starch, glycogen, cellulose and pectin.

Classification by function: storage polysaccharides (starch in plants, glycogen in animals) and structural polysaccharides (cellulose).

Starch is the main storage of energy in plants and consists of glucose units. It has two components. **Amylose** has long, unbranched chains with many (1,4) glycosidic linkages. **Amylopectin** is a compact, highly branched structure of many glucose units (1,4 links with 1,6 links at the branches). According to the textbook, amylopectin is soluble in water while amylose is insoluble. Potato, sugar beet, rice and grains contain large quantities of starch.

Cellulose is the major structural material of plant cell walls. It consists of glucose units (up to 2500) forming long unbranched chains linked by glycosidic bonds. These chains form strong hydrogen bonds with each other, making cellulose rigid and insoluble in water. It makes up most of the leaves, branches and trunk of plants. It is very important in human food in the form of fibre, which improves gut health.

Glycogen, or 'animal starch', is a polymer of glucose that is stored in the liver and muscles of animals. Its structure is very similar to that of amylopectin, but it is more highly branched, which makes it different from starch.

Polysaccharide	Structure	Function
Starch – amylose	Long, unbranched chains of glucose	Energy store of plants
Starch – amylopectin	Compact, highly branched chains of glucose	Energy store of plants
Cellulose	Long unbranched chains (up to 2500 glucose units) held together by strong hydrogen bonds	Structural material of plant cell walls; dietary fibre
Glycogen	Like amylopectin but more highly branched	Energy store of animals (liver and muscles)

Q5. How the structure of proteins support its function. Describe at each structure level.

Primary structure: the specific linear sequence of amino acids linked by peptide bonds in a polypeptide chain. This sequence determines how the protein folds into its three-dimensional shape, which in turn defines its biological function. Even a small change in a single amino acid can alter the structure and impair the function. Proteins like albumin (about 400 amino acids) and insulin rely on their precise amino acid sequences to function properly.

Secondary structure: it arises when the backbone atoms form hydrogen bonds within a polypeptide chain, producing regular patterns such as α-helices and β-pleated sheets. In an **α-helix**, hydrogen bonds form along a single chain, creating a coiled, spiral structure (e.g. keratin). **β-pleated sheets** form when hydrogen bonds occur between adjacent polypeptide strands, giving a folded, sheet-like structure (e.g. silk fibroin). These arrangements provide stability and flexibility.

Tertiary structure: the overall three-dimensional shape formed when a polypeptide chain folds due to interactions between the side chains (R-groups): hydrogen bonds, ionic bonds, hydrophobic interactions and disulphide bridges. These interactions make the chain twist and bend into a specific shape, often forming globular proteins. This three-dimensional structure determines the function of the protein. For example, in enzymes the precise shape of the active site allows specific binding of the substrate; any change in this structure can affect the enzyme activity.

Quaternary structure: some proteins have it too. It arises when two or more folded polypeptide chains (subunits) associate by interactions such as hydrogen bonds, ionic (salt) bridges, hydrophobic interactions and sometimes disulphide bonds.

Level	What it is	Bonds involved	Example / function
Primary	Linear sequence of amino acids	Peptide bonds	Decides folding and function (albumin, insulin)
Secondary	α-helix or β-pleated sheet	Hydrogen bonds of the backbone	Keratin (helix), silk fibroin (sheet)
Tertiary	Overall 3-D folding of one chain	H-bonds, ionic bonds, hydrophobic interactions, disulphide bridges	Active site of enzymes
Quaternary	Two or more chains (subunits) together	Same interactions between subunits	Proteins with several subunits

Q6. Write an account of classification of lipids with suitable examples.

Based on their chemical composition, lipids are classified into simple lipids, compound lipids and derived lipids.

Simple lipids are esters of fatty acids with different types of alcohols; they are sub-classified by the type of alcohol into fats, oils and waxes.

Triglycerides: animal fats and vegetable oils are esters formed from one glycerol molecule and three fatty acids, which may be similar or different. This structure makes them efficient energy storage molecules; they also provide insulation, protection of organs and a source of fat-soluble vitamins.

$$CH_2OH–CHOH–CH_2OH + 3\ R–COOH \rightarrow CH_2(OOCR)–CH(OOCR)–CH_2(OOCR) + 3\ H_2O$$

glycerol + three fatty acids → triglyceride + water

Most waxes are made up of long fatty acid chains esterified to long-chain alcohols. Both chains are non-polar and hydrophobic. Because of this, waxes serve as barriers by forming protective coatings which prevent water from sticking to the surface. Natural examples include beeswax, the oil in a whale's head, the feathers of some aquatic birds and plant waxes. They are widely used in the pharmaceutical, cosmetic and other industries to make lotions, ointments and polishes.

Compound lipids are esters of fatty acids with alcohols containing an additional group, such as phosphate, nitrogenous base, carbohydrate or protein.

Phospholipids: similar to a fat molecule but with only two fatty acids attached to glycerol; the third hydroxyl group is linked to a phosphate group, which makes the molecule polar. The hydrocarbon tails are hydrophobic, while the phosphate group and its attached components form a hydrophilic head. In water, phospholipids form a double-layered sheet called a **bilayer**: the polar ends face outward and the non-polar ends cluster together inside. It forms the structure of all cell membranes.

Lipoproteins: complexes of lipids and proteins that transport hydrophobic lipids (like cholesterol and triglycerides) in the bloodstream.

Glycolipids: lipids with carbohydrate (sugar) groups attached, mostly found in nervous tissue.

Derived (associated) lipids are the hydrolytic products of simple and compound lipids which have the same characteristics as lipids, e.g. fatty acids, glycerol, steroids and lipid-soluble vitamins. They are not fats themselves, but come from fats. They have important roles in energy and hormone production, cell structure, nerve health and vitamin function.

Steroids are not formed from fatty acids but have some lipid-like properties. They consist of three cyclohexane rings and one cyclopentane ring fused together, called the **steroid nucleus**; the four rings are named A, B, C and D. Many steroids also have the –OH functional group, which puts them in the alcohol class (sterols). The steroids of biological systems include cholesterol, bile salts, vitamin D and steroid hormones.

Lipids present in plants include a wide variety of hydrocarbons called terpenes, which are built from isoprene units. These hydrocarbons and their oxygenated derivatives have fewer than 40 carbon atoms. Terpenes that contain 40 carbon atoms are known as **carotenoids**. Example: limonene is a terpene in lemon.

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Q7. What are nucleic acids? Describe the roll of DNA and RNA in the protein synthesis.

Nucleic acids are large biomolecules that store and transmit genetic information. They are polymers made up of smaller units called **nucleotides**. They are formed from nucleotide chains containing alternating sugar and phosphate groups, with organic bases attached to the sugar molecules. The sugars are ribose in RNA and deoxyribose in DNA; both are five-membered sugars.

Nucleotides are joined together by a **phosphodiester linkage** between the 5′ and 3′ carbon atoms of the sugars. The process is repeated up to several million times to make a polynucleotide. Phosphodiester linkages are strong covalent bonds, which give strength and stability to the polynucleotide chain.

Role of DNA: DNA is the coded plan for making proteins. Each protein is made by a specific gene; the sequence of bases in the gene (three bases for each amino acid) fixes the sequence of amino acids.

Role of RNA: mRNA carries the copy of the gene to the ribosome, rRNA helps to make up the ribosome, and tRNA translates the base sequence into the sequence of amino acids.

Transcription: the genetic instructions for making a protein are copied from a DNA gene into mRNA in the nucleus. When the mRNA strand is complete, it detaches from DNA and enters the cytoplasm, where it attaches to ribosomes.

Translation: at the ribosomes, the code carried by mRNA is read and converted into a specific sequence of amino acids. Amino acids are arranged according to the sequence of bases in the RNA. The tRNA molecules release their amino acids, which join together in a chain; the polypeptide chain grows as bonds form between the amino acids, and finally the complete protein molecule is formed.

DNA → (transcription) → mRNA → (translation) → protein

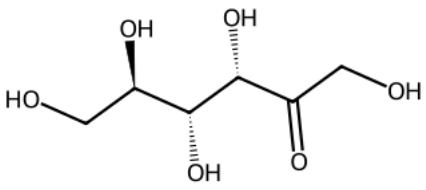
QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the “Quick Check” boxes inside the chapter. The textbook numbers two boxes as Quick Check 27.3 (enzymes and lipids); both are solved in the printed order.

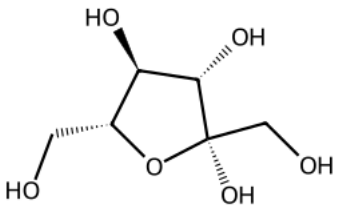
Quick Check 27.1

a) The linear structure of fructose is given here. i. Show the conversion of this structure into the cyclic structure. ii. Predict whether fructose is soluble in water or not. Explain your reasoning. iii. Predict the physical state of fructose in its pure form by analyzing its structure.

i. The –OH group on carbon 5 adds to the ketonic C=O group on carbon 2, closing a five-membered ring (furanose).



fructose: open-chain form (ketose)



fructose: five-membered ring (furanose)

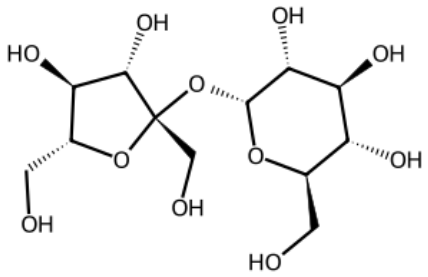
ii. Fructose is soluble in water. It has many hydroxyl (–OH) groups which form extensive hydrogen bonds with water molecules.

iii. It is a solid in its pure form, because the many –OH groups give extensive hydrogen bonding between its molecules.

b) What is glycosidic linkage? Show how it is formed between fructose and glucose.

The C–O–C bond which joins two monosaccharide units is called the glycosidic linkage. It is formed by a condensation reaction: the hydroxyl group of glucose combines with the hydrogen of a hydroxyl group of fructose, a molecule of water is released and the two rings are joined through an oxygen atom to give sucrose.

$$C_6H_{12}O_6 \text{ (glucose)} + C_6H_{12}O_6 \text{ (fructose)} \rightarrow C_{12}H_{22}O_{11} \text{ (sucrose)} + H_2O$$



sucrose: glucose ring and fructose ring joined by a C–O–C glycosidic linkage

c) Briefly describe any two functions of carbohydrates and how these functions are related to their molecular structures.

1. Energy storage: starch (plants) and glycogen (animals) store energy. They are polymers of glucose; their branched, compact structures pack many glucose units, which are released on hydrolysis.

2. Structural support: cellulose forms plant cell walls. Its long unbranched chains of glucose form strong hydrogen bonds with each other, making it rigid and insoluble in water.

Quick Check 27.2

a) Keratin is a structural protein found in hair, nails, horn, hoofs, wool, feathers, enamel of teeth and outer layer of skin, where it gives strength and flexibility. It is not easily denatured. Which bond do you expect is responsible for this; hydrogen bond or disulphide bond? Explain.

The **disulphide bond**. It is a strong covalent bond (–S–S–) between the chains and is not easily broken by heat or a change of pH, whereas hydrogen bonds are weak and are easily disrupted. So the disulphide bridges keep the keratin chains firmly linked.

b) How can two peptides with identical type and number of amino acids differ in their primary structure?

The primary structure is the specific linear **sequence** of amino acids. The same amino acids can be joined in a different order (e.g. Gly–Ala–Ser and Ala–Gly–Ser), which gives different primary structures and therefore different peptides.

c) What are the two most common types of secondary protein structures in our body?

The α-helix and the β-pleated sheet.

d) Why does a denatured protein no longer function properly?

The function of a protein depends on its specific three-dimensional shape. On denaturation the hydrogen bonds (and other interactions) that hold this shape are disrupted and the chains open up, so the shape (e.g. the active site of an enzyme) is lost and the protein cannot work.

e) The white of egg solidifies on heating. Explain why it happens.

Egg white is a globular protein. On heating, the hydrogen bonding between the chains is disrupted and the chains open up, so the protein is denatured and hardens (coagulates).

Quick Check 27.3

a) How do enzymes differ from inorganic catalysts?

Enzymes are proteins (biological catalysts) and are highly specific: each acts on only one particular substrate, which fits its active site. They work best only at an optimum temperature (about 37 °C) and pH, and are deactivated outside these. Inorganic catalysts are not proteins, are less specific and can work over a wide range of temperature and pH.

b) What is the effect of decreasing and increasing temperature affect an enzyme activity?

Enzyme activity is maximum at the optimum temperature (about 37 °C in the human body). On decreasing the temperature the molecules move more slowly, fewer enzyme–substrate complexes form and the activity falls. On increasing the temperature above the optimum the enzyme (a protein) is denatured and its action is deactivated.

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c) Differentiate between competitive and non-competitive inhibitors.

Competitive inhibition	Non-competitive inhibition
The inhibitor has the same shape as the substrate.	The inhibitor is a small ion or molecule that need not resemble the substrate.
It occupies the active site and competes with the substrate.	It changes the shape of the active site so that the substrate cannot bind.
Reversible.	Irreversible; activity is lost permanently.
Example: malonate inhibits succinate dehydrogenase.	Example: penicillin blocks trans-peptidase; poisons.

d) An enzyme is a protein. When the pH is changed, the function of the enzyme is disturbed. In terms of the protein structure, explain how and why it happens so?

The active site has a precise shape which is part of the tertiary structure, held by hydrogen bonds and ionic bonds between the R-groups. A change in pH changes the charges on these groups and disrupts these bonds. The chain unfolds (denatures), the shape of the active site changes and the substrate can no longer bind.

Quick Check 27.3 (second box, lipids)

a) Why lipids are not soluble in water?

Because of the long, non-polar (hydrophobic) hydrocarbon chains of their fatty acids, which cannot form hydrogen bonds with water. They dissolve in organic solvents instead.

b) What are three major functions of lipids in the human body?

(1) They are a primary store of energy. (2) They act as solvents for the fat-soluble vitamins (A, D, E and K) and help in their absorption. (3) They insulate the body and cushion organs such as the kidneys (phospholipids also form cell membranes).

c) What type/s of lipid do/does not contain fatty acids? Give examples.

Steroids and terpenes are not formed from fatty acids. Examples: cholesterol, bile salts, vitamin D and steroid hormones (steroids); limonene and carotenoids (terpenes).

d) Vegetable oil and vegetable ghee both are triglycerides. What is the major structural difference between the two?

Vegetable oil contains unsaturated fatty acids with one or more C=C double bonds and is a liquid. Vegetable ghee contains saturated fatty acids (no C=C double bonds) and is a solid at room temperature.

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Quick Check 27.4

a) Compare the stability provided by the sugar-phosphate backbone versus the inter-chain base pairing. In your answer, refer to the specific types of bonds and the number of bonds formed between: Adenine (A) and Thymine (T); Guanine (G) and Cytosine (C).

The sugar–phosphate backbone is held by **phosphodiester linkages**, which are strong covalent bonds; they give strength and stability to each polynucleotide chain. The two chains are held to each other by **hydrogen bonds** between the bases, which are weak: **two** hydrogen bonds between A and T and **three** between G and C. So the backbone gives the greater stability, while the weak base pairing allows the strands to separate for replication and transcription.

b) How the DNA and RNA are chemically different? What are the similarities between these two?

DNA	RNA
Sugar is deoxyribose.	Sugar is ribose.
Bases: A, T, G and C.	Bases: A, U, G and C (uracil in place of thymine).
Double stranded (double helix).	Usually single stranded; can fold back into a helical loop.
Stores the genetic information.	Transfers the information for protein synthesis (mRNA, rRNA, tRNA).

Similarities: both are polymers of nucleotides; both have a sugar–phosphate backbone joined by phosphodiester linkages; both contain the bases A, G and C; both carry genetic information.

c) Nearly every cell in the human body contains the exact same 23 pairs of chromosomes, yet a red blood cell produces haemoglobin while a skin cell produces keratin. Explain why.

Each cell activates only the genes it needs and keeps the others switched off. A red blood cell activates the gene for haemoglobin while keeping the gene for keratin switched off; in a skin cell the keratin gene is active.